Automated Detection of Satellite Trails in Ground-Based Observations Using U-Net and the Probabilistic Hough Transform

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July 2026

Abstract

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The proliferation of low-Earth-orbit satellite constellations has made automated detection and masking of satellite trails a routine requirement for wide-field observatories. This dissertation replicates the U-Net plus probabilistic Hough transform pipeline of Stoppa et al. [28] on a labelled MeerLICHT subset, reimplemented in PyTorch with an architecture-faithful network, a combined binary cross-entropy and squared-Dice loss, validation-only model and threshold selection, and Hough post-processing of the probability map. The replication is then examined through four analyses: residual error, model variants, training-protocol sensitivity, and inference-time strategies. It reproduces the paper's recall (five-seed test recall 0.923 against 0.94) and its Hough completeness (trail-pixel recall $0.918 \rightarrow 0.990$; a pixel-level false-negative rate $0.082 \rightarrow 0.0105$ comparable to the paper's $6.98\,\% \rightarrow 3.38\,\%$), while strict pixel precision sits below the paper and is prevalence-dependent. The residual-error analysis localises the gap: 82.5 % of false positives are intra-patch, a one-pixel tolerance raises five-seed precision from 0.847 to 0.946, and 57 % of false positives lie within one pixel of the mask — consistent with error dominated by boundary placement rather than spurious detection. The model, training, and inference variants move the residual only slightly, so on this dataset the remaining gap is unlikely to stem from model capacity, training protocol, or Hough post-processing alone.

Contents

Total LaTeX Word Count: 6950/7000 (excluding Contents, Figures, Tables, Bibliography and Appendix)

Chapter 1

Introduction and Context

The population of low-Earth orbit is growing quickly. CelesTrak listed 15,848 active satellites on 27 May 2026, of which approximately 14,600 occupied orbits below 2,000 km, including more than 10,000 Starlink spacecraft [7]. Authorisations extend further: 15,000 second-generation Starlink satellites are approved [10], and policy reviews treat proposals exceeding 100,000 spacecraft as realistic [34, 35]. Each sunlit satellite crossing a wide-field exposure leaves a bright, near-linear trail that corrupts photometry and astrometry along its path and injects spurious candidates into transient-detection pipelines [11, 31].

The contamination is already measurable. Starlink trails affected under 0.5 % of Zwicky Transient Facility twilight images in late 2019 but 18 % by August 2021 [20]; Rubin Observatory simulations indicate that approximately 10 % of LSST images would contain at least one trail under a 40,000-satellite scenario [32]; wide-field twilight exposures can reach 30–40 % contamination [2, 11]; and 4.3 ± 0.4 % of Hubble ACS/WFC images taken in 2018–2021 contained trails [16]. At these rates manual flagging does not scale: a survey instrument such as MeerLICHT [4] produces $10{,}560 \times 10{,}560$-pixel frames throughout the night, so identifying and masking affected pixels must be automated, reliable at the pixel level, and cheap enough to run inside the reduction pipeline.

1.1 The target pipeline

Stoppa et al. [28] present ASTA, the pipeline replicated in this dissertation: a U-Net [24] trained on manually annotated MeerLICHT images, followed by a probabilistic Hough transform [14, 17] applied to the network's output probability map. The division of labour is deliberate. The U-Net supplies per-pixel soft detection that survives the crowded stellar background, and the Hough stage exploits the one prior the network does not encode explicitly, that a satellite trail is globally linear, to connect fragmented detections and extend partially recovered trails. The published pipeline reaches precision and recall both approaching 0.94 and was applied at scale to around 200,000 full MeerLICHT fields; its reported behaviour is detailed in Section 1.2.3, and this work's numbers are deferred to Chapter 5.

1.2 Background and related work

1.2.1 Classical trail detection

Classical satellite-trail detection usually treats trails as geometric line features. The Hough transform [14], its bounded (ρ, θ) parameterisation [9], and the progressive probabilistic variant for noisy segments [17] map image pixels into line-parameter space, making them natural tools for long, straight artefacts. Astronomical applications include Hough-style voting and renewal strings for survey cleaning [29], masked-star Hough validation in crowded fields [3], Radon-style matched filtering for faint streaks [21], space-debris and DECam survey workflows [25, 33], and modified Radon recovery in Hubble ACS imaging [27]. Their shared limitation is sensitivity to faint, fragmented, or background-confused trails, where reliable voting and instrument-specific preprocessing become difficult; this motivates learned segmentation before geometric post-processing.

1.2.2 Deep learning for astronomical segmentation

The U-Net architecture [24], an encoder–decoder with skip connections that preserve spatial resolution, was designed for biomedical segmentation and is now common in astronomical imaging. Close precedents for thin, sparse artefacts are deepCR [38] and Cosmic-CoNN [36], both cosmic-ray segmentation systems. MaxiMask [23] covers contaminant masking including satellite and aircraft trails, and trails have since been targeted with U-Net plus line-segment detection [8], multi-band attention U-Nets [37], end-to-end streak cataloguing [5], and frequency-domain alert-stream classification [6]. No published trail-detection pipeline was found that gates a segmentation U-Net with a lightweight patch classifier, although deep-learning classifiers already operate elsewhere in the MeerLICHT ecosystem [13]. Section 5.7.1 therefore treats this as a narrow follow-up test.

1.2.3 The target pipeline: Stoppa et al. (2024)

Stoppa et al. [28] trained a U-Net on 528×528 MeerLICHT patches with a combined binary cross-entropy and Dice loss, and applied a probabilistic Hough transform to the U-Net probability map as a post-processing step. On a 20,000-patch test set at threshold 0.58 the pipeline reported precision and recall both approaching 0.94, with the Hough step reducing the pixel-level false-negative rate from 6.98 % to 3.38 %. The paper does not report the training hyperparameters, making faithful replication a non-trivial search (Chapter 4).

1.3 Aims, scope, and contributions

This dissertation has three aims. The first is a faithful PyTorch replication of the Stoppa et al. [28] pipeline on the labelled MeerLICHT subset, with the network architecture recovered from the released model, not the paper's text alone, and with model and threshold selection locked on validation data before any test metric is inspected. The second is to diagnose the residual error of that replication: to establish where the strict precision gap against the published figures comes from, rather than to report it as a single number. The third is to test which interventions can move that residual, through model variants, training-protocol changes, and inference-time strategies.

The contributions are:

- an architecture-faithful, validation-locked replication that reproduces the paper's recall and Hough pixel completeness (Chapter 5);
- a residual-error characterisation that localises the strict precision gap at the one-pixel boundary scale;
- controlled tests of a classifier-gated detector, an attention-gated U-Net, hard-negative mining, soft labels, ensembling, and probability-stratified Hough, each interpreted against that diagnosis;
- a reproducibility layer of preregistered selection rules, multi-seed reporting, and packaged, tested, documented code (Section 4.10).

Sky-coordinate projection, horizon and zenith-angle analysis, and exposure-time-normalised trend modelling are out of scope: the available data are 8-bit PNG renders with no surviving FITS or WCS headers (Section 2.1).

1.4 Code and data availability

The source code, configuration files, unit tests, and the locked model checkpoint are available in the GitHub repository, with rendered API documentation on Read the Docs. The MeerLICHT image and mask subset is collaboration data, provided through the supervisor under the MeerLICHT data policy, and is not redistributed; the splits are reproducible from `src/data/splits.py` (seed 2804). The DECam frames used in Section 5.5 are public NSF NOIRLab Astro Data Archive products, identified by exposure and detector number in `results/classical/decam_cold_manifest.json`. All training and evaluation runs used the Cambridge Service for Data Driven Discovery (CSD3), operated by the University of Cambridge Research Computing Service and funded by EPSRC Tier-2 capital grant EP/T022159/1 and DiRAC funding from the STFC.

Chapter 2

Data Exploration

2.1 The MeerLICHT subset

MeerLICHT is a dedicated 65 cm wide-field optical telescope in Sutherland, South Africa, designed to follow the radio sky in the optical band [4]. Its 10,560 $\times$ 10,560-pixel CCD samples a 2.7 deg^2 field of view at $0.56''$ pixel^{-1}, so a 528 $\times$ 528 patch subtends about $4.9'$ on a side. The labelled subset available for this work comprises 178 full-frame image–mask pairs: 8-bit PNG renders of FITS exposures, each paired with a hand-annotated binary trail mask (Figure 2.1). The renders carry no surviving FITS, WCS, exposure, or photometric-calibration headers, which caps the per-pixel dynamic range and precludes the sky-coordinate, airmass, and exposure-time analyses noted in Section 1.3. The data are therefore treated as a labelled display-space segmentation corpus, not as calibrated astronomical images.

2.2 Exploratory analysis

Exploratory analysis characterises the trail population and the per-image trail burden the splits must balance. Annotated trails are long, near-linear features: connected components have small minor-axis widths and high aspect ratios, and many trails cross several adjacent patches. The per-image trail-pixel count is heavily skewed, which motivates the stratified split of Section 2.4 rather than a random split over images. Filename-derived observation dates span 2017–2023, dominated by 2022 exposures; this is context only and carries no time-trend claim, since the subset is neither exposure-time nor observing-condition normalised.

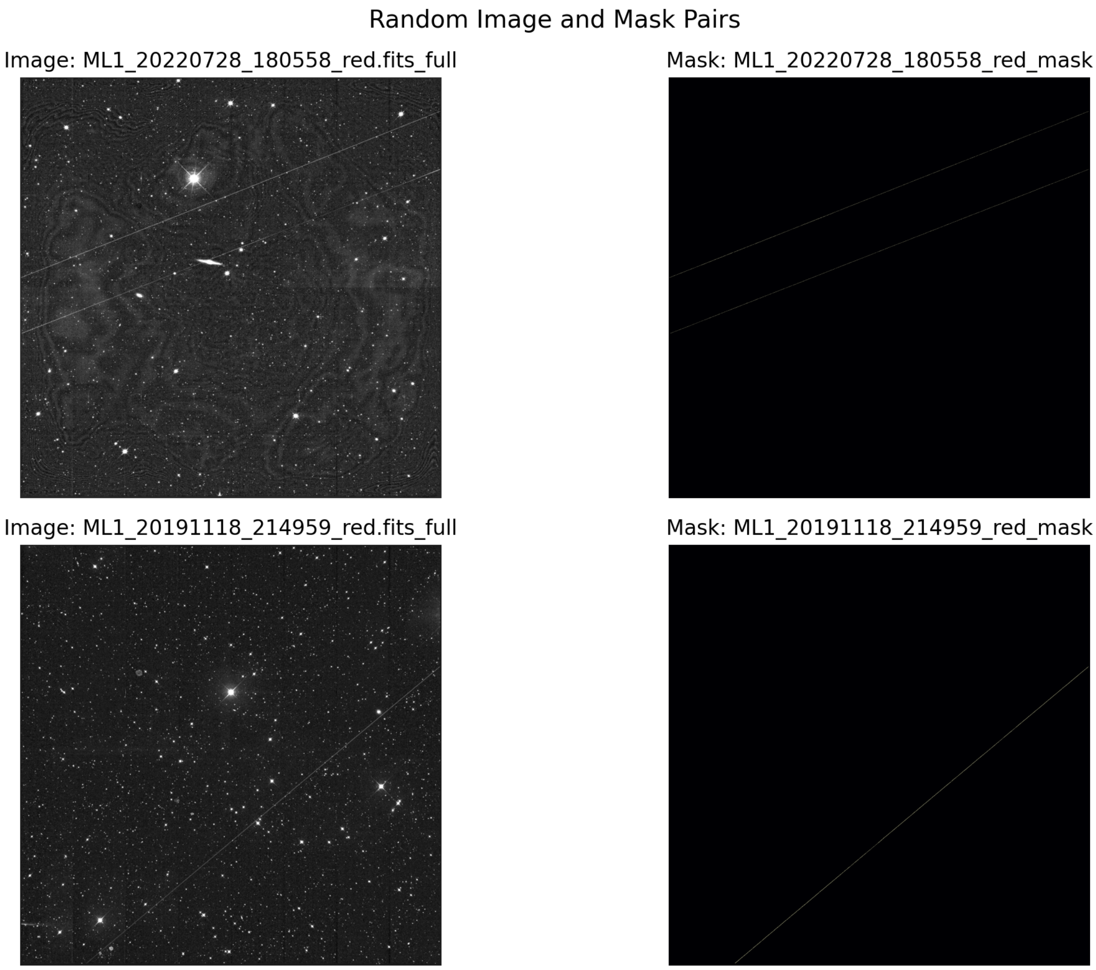

Figure 2.1: Example image–mask pairs from the labelled subset: 8-bit PNG renders with hand-annotated binary trail masks. Trails are long, near-linear features against a crowded stellar background.

2.3 Patch extraction and class imbalance

Full frames are too large for direct training. Patches are extracted at 528×528 pixels with stride 528, yielding exactly 400 non-overlapping patches per image (20×20 grid). The natural patch distribution is highly imbalanced, with approximately 93–94 % of full-grid patches containing no trail pixels, so the sampled manifests retain all positive patches and draw three negatives per positive. Figure 2.2 gives the realised 528-pixel manifest sizes used by the final architecture-faithful run.

The sampled test set is the internally consistent evaluation population, because it matches the sampled validation prior used for threshold selection. A separate all-negative parity manifest keeps all 12,800 test patches (872 positive, 11,928 negative) as a paper-facing check (Figure 2.2); it is never used for model or threshold selection.

2.4 Dataset splits

Figure 2.2 summarises the path from frames to evaluation manifests. Splits are performed at the image level, never the patch level, so no two patches from the same frame can fall in different splits. The 178 images are stratified by total trail-pixel count into four quartile groups, and the nominal 0.70/0.15/0.15 ratios are applied within each group with flooring; the remainder goes to test. The realised split is therefore **122 train / 24 validation / 32 test** (0.685/0.135/0.180), quoted throughout. This is leakage-safe and conservative, reserving a relatively large held-out fraction for evaluation, and splits are seeded (seed 2804).

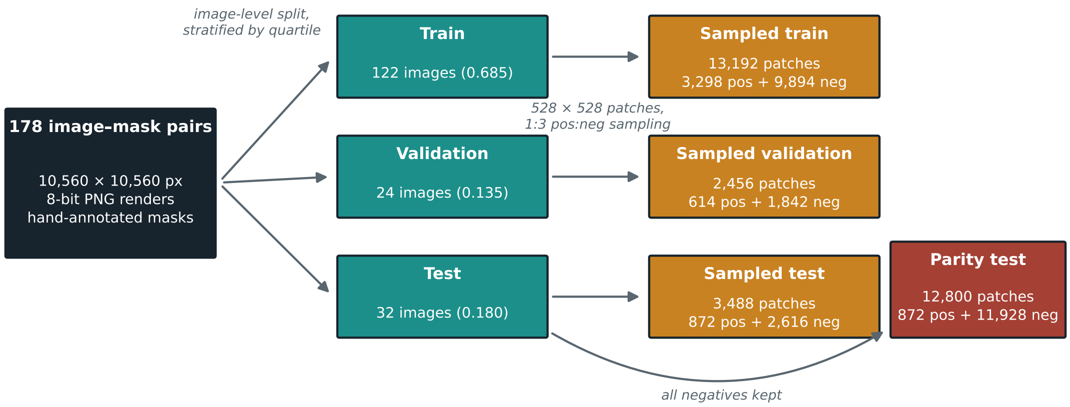

Figure 2.2: Data flow from the 178 labelled frames to the evaluation manifests. Sampled manifests drive training, threshold selection, and internal evaluation; the parity manifest restores all negatives for the paper-facing precision check and is never used for selection.

2.5 Mask quality

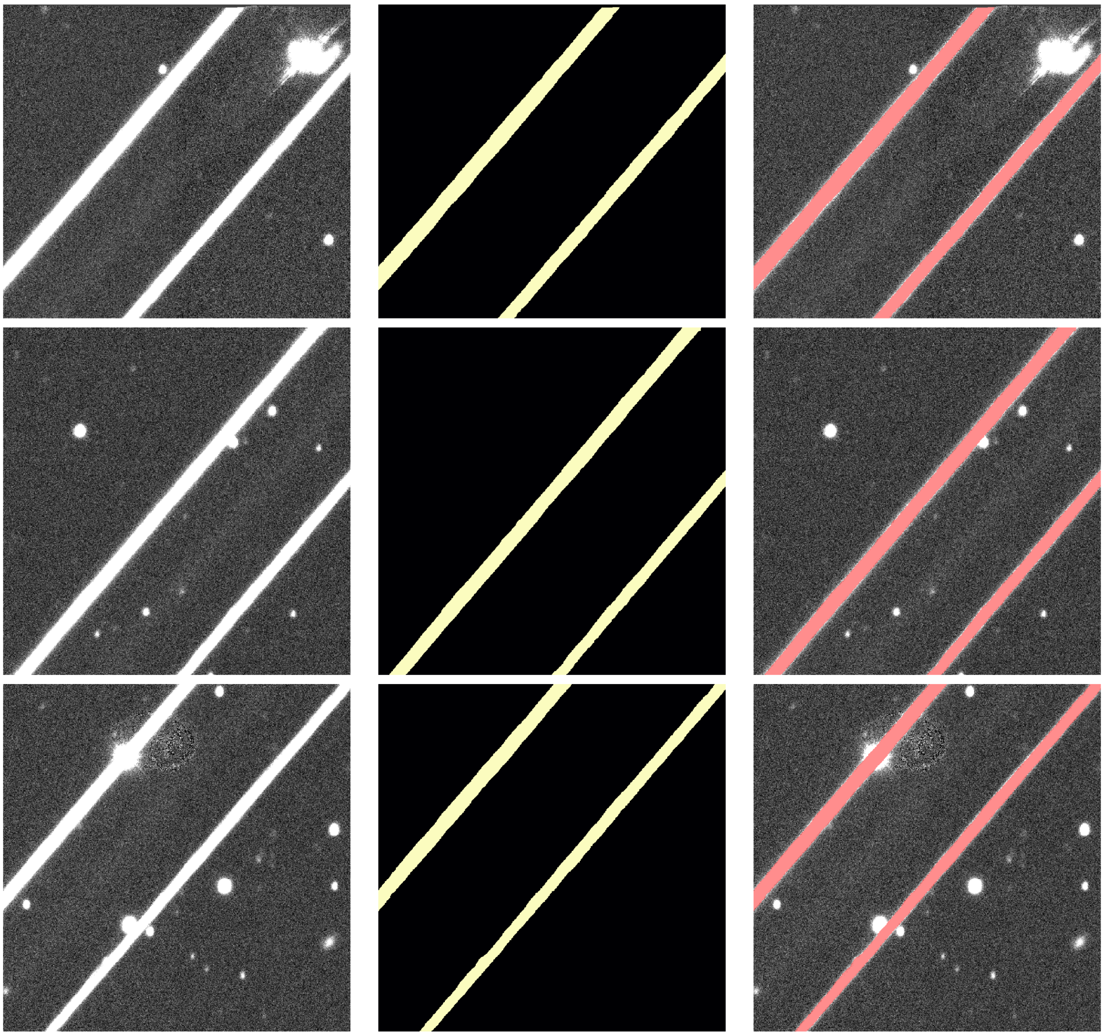

Figure 2.3: Mask-quality inspection grid for three of the densest patches: image, annotated mask, and overlay (red). The masks track the visible trails; the residual concern is undercoverage at faint endpoints, quantified in Section 5.6.2.

Connected-component analysis is a mask-quality check: components have high aspect ratios and a median stroke width near 6 px (distance transform), confirming genuinely linear structures, and the densest patches adequately cover the visible trails (Figure 2.3).

The masks are sufficient for supervised training, with one characterised caveat: full-frame overlays show occasional undercoverage at faint endpoints and low-contrast stretches. A model that detects a genuine but unlabelled extension is penalised as a false positive, so strict label-referenced precision is a lower bound where labels undercover. Section 5.6.2 quantifies this boundary disagreement; interpretations are therefore phrased as 'consistent with' boundary disagreement, not as proof of label error.

Chapter 3

Methodology

3.1 Problem formulation

Trail detection is cast as patch-wise binary semantic segmentation: each 528×528 single-channel patch is mapped to a per-pixel trail/background probability map. This matches the practical masking problem, since downstream processing needs to know which pixels are contaminated, not which satellite produced them. The framing is semantic rather than instance-level, so crossing trails within a patch are all simply "trail"; the pixel-level metrics therefore cannot distinguish merged from separated trails, a blind spot revisited in Section 6.4.

3.2 Why learned segmentation with geometric post-processing

The pipeline pairs a learned segmentation stage with a classical geometric stage. A U-Net produces a soft-detection map that preserves the weak, spatially extended evidence of a faint trail. A classical Hough detector applied directly to the raw image responds mainly to stars, galaxies, and diffraction structure; in the preliminary baseline it produced precision of order 10^{-3}, unusable for masking. The probabilistic Hough transform is therefore used after the U-Net, where the input is already a trail-probability canvas rather than a crowded sky image, and where it can exploit the prior that a trail is globally linear to recover coherent extent across gaps in the probability map.

3.3 Why a U-Net

The U-Net encoder–decoder with skip connections [24] preserves full spatial resolution, essential for thin, elongated, sparse targets. A trail must be localised at the pixel boundary while still drawing on enough context to distinguish it from stellar structure or background gradients. The same architecture underpins the closest astronomical artefact detectors, deepCR [38] and MaxiMask [23], supporting its suitability here. It is also the model family used by the target paper, so a U-Net is required for a controlled replication.

3.4 Why a combined BCE and Dice loss

The foreground occupies a tiny fraction of pixels, so a pure cross-entropy objective is dominated by easy background. The combined loss adds a Dice term [19], which measures region overlap and is insensitive to the background majority, to the per-pixel calibration of binary cross-entropy, a Combo-loss pairing [30] designed for severely imbalanced segmentation. No additional positive `pos_weight` is applied in the final pipeline: the 1:3 patch sampling and Dice term already address the imbalance, while stacking another class weight would bias the model towards over-prediction. The exact formulation is given in Section 4.3.

3.5 Evaluation philosophy

Four principles govern the evaluation. Model and threshold selection are pre-registered and validation-only, so no test metric is inspected before the configuration is locked. Sampled and all-negative test metrics are kept distinct: the sampled set matches the validation prior, while the all-negative set restores the natural empty-patch prevalence and is closer to the target paper. Uncertainty treats the source image, not the patch, as the independent unit, because patches from one frame share background, seeing, and annotation structure. Finally, because the labels undercover (Section 2.5), strict pixel-overlap scores are complemented by topology-aware (clDice), false-positive distance, and boundary-tolerant metrics (Section 4.8).

Chapter 4

Training, Architecture, and Data Pipeline

4.1 Data pipeline

The on-disk patch builder extracts 528×528 patches at stride 528 (Section 2.3), writing canonical orientations at the 1:3 sampling ratio. The builder records the source image, split, patch coordinates, positive-pixel fraction, and full-image display-space mean and standard deviation in the manifest. The last two fields implement the `full_image` normalisation mode: each patch is converted to $[0, 1]$ and normalised with the statistics of its parent image, matching the paper's per-image standardisation better than a per-patch z-score.

4.1.1 Data augmentation

Training patches receive random horizontal flips, vertical flips, and $k \times 90°$ rotations applied identically to image and mask at load time, so stored patches keep their canonical orientation and can appear in different orientations across epochs; validation and test patches are not geometrically augmented. Training images also receive signal-dependent stochastic noise (images only, never masks or evaluation splits). For a patch-pixel value $p \in [0, 1]$ the augmentation samples

$$p_{\text{noisy}} = \text{clip}\Big(p + \mathcal{N}\Big(0,\ m\sqrt{\alpha p + \beta^2}\Big)\,,\ 0,\ 1\Big)\,,$$

where m is the noise multiplier and α, β are calibrated once from empty-mask training patches by `scripts/data/compute_background_noise_stats.py`, so the injected noise matches the measured background variation. The locked calibration uses 9,894 empty train patches, giving $\alpha = 0.0209$ and $\beta = 0.0886$. It is described as signal-dependent stochastic noise rather than Poisson–Gaussian CCD noise because the inputs are display renders, not calibrated electron counts. The active configuration fixes $m = 1.0$ as a calibrated, robustness-motivated choice rather than a tuned gain; a preliminary three-seed screen over multipliers $\{0.5, 1.0, 2.0\}$ found no gain meeting the preregistered adoption criterion.

4.2 Network architecture

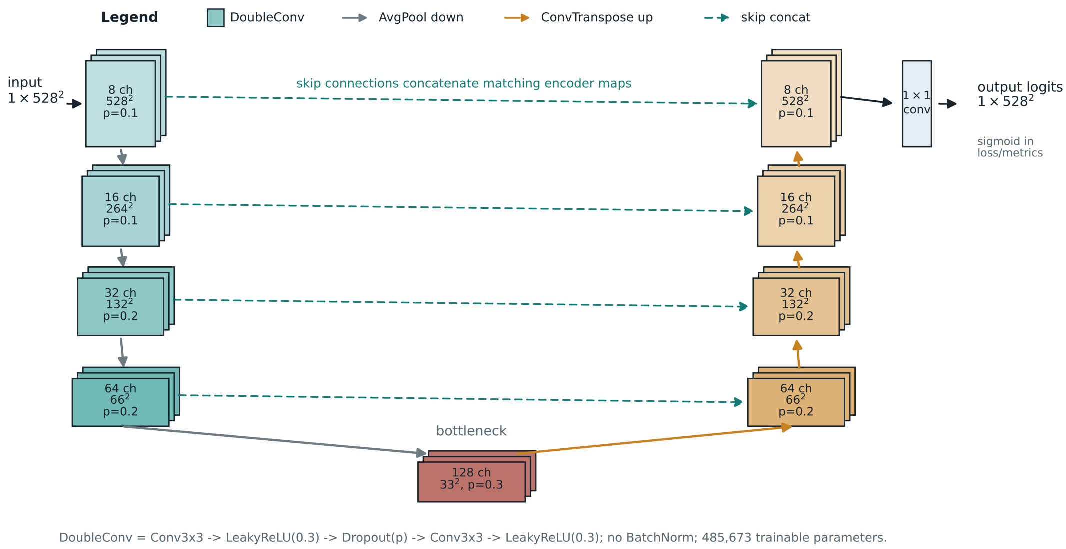

Figure 4.1: Architecture-faithful U-Net. Encoder widths $8 \to 16 \to 32 \to 64$ with a 128-channel bottleneck, average-pooling downsampling, transposed-convolution upsampling, and skip concatenation at each level. Each block is two 3×3 convolutions with LeakyReLU(0.3) and in-block dropout.

The replication network (Figure 4.1) follows the released Stoppa et al. model configuration rather than only the paper's textual description: an encoder–decoder with skip connections and filters progressing $8 \to 16 \to 32 \to 64 \to 128$, average-pooling downsampling, transposed-convolution upsampling, and LeakyReLU activations with slope 0.3. Each double-convolution block follows `Conv2d` $\to$ `LeakyReLU` $\to$ `Dropout` $\to$ `Conv2d` $\to$ `LeakyReLU`, with element-wise dropout in all nine blocks. The dropout rates from stem to final decoder block are

$$0.1,\ 0.1,\ 0.2,\ 0.2,\ 0.3,\ 0.2,\ 0.2,\ 0.1,\ 0.1.$$

The network uses no batch normalisation. The 3×3 convolutions use He-normal initialisation [12]; the transposed convolutions and the final 1×1 head use Glorot-uniform initialisation; all biases are initialised to zero. Inputs are 528×528 single-channel patches and the output is a single-channel logit map of identical spatial dimensions: the model returns logits, and the sigmoid is applied explicitly in the loss, threshold, Hough, and visualisation paths. The network has 485,673 trainable parameters (`base_channels` $= 8$).

4.3 Loss function

Training uses a combined binary cross-entropy (BCE) and squared-Dice loss:

$$\mathcal{L} = \lambda_{\mathrm{BCE}}\,\mathcal{L}_{\mathrm{BCE}} + \lambda_{\mathrm{Dice}}\,\mathcal{L}_{\mathrm{Dice}}, \qquad \lambda_{\mathrm{Dice}} = 1 - \lambda_{\mathrm{BCE}}, \tag{4.1}$$

where

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i^2 + \sum_i g_i^2 + \epsilon}, \qquad \epsilon = 10^{-4},$$

with p_i the sigmoid probabilities and g_i the binary targets. The Dice term follows Milletari et al. [19]; combined with BCE it sits in the Combo-loss family of Taghanaki et al. [30]. The BCE term uses `BCEWithLogitsLoss` without a class-balancing `pos_weight`, matching the paper-faithful path and avoiding a second imbalance correction on top of the sampled patch prior. Losses are evaluated in FP32 even when the forward pass uses BF16 autocast [15, 18].

4.4 Hyperparameter search

Hyperparameter search uses Optuna [1] with a tree-structured Parzen estimator and median pruner ($n_{\mathrm{startup}} = 15$, 10 warm-up epochs). Forty-five trials are balanced across batch sizes $\{8, 16, 32\}$ (fifteen each) after a preliminary free-sampling sweep concentrated most of its trials on one batch size and pruned large-batch trials under a higher conditional learning-rate ceiling. The searched continuous hyperparameters are learning rate (log-uniform in $[10^{-4}, 10^{-3}]$) and BCE weight λ_{BCE} (uniform in $[0.2, 0.8]$); dropout, normalisation, and noise multiplier are fixed. Each trial runs for 20 epochs and is scored by threshold-swept validation F_1, matching the final selection criterion. Pruning uses in-epoch validation Dice at the fixed training threshold as an efficiency heuristic, while only completed trials are rescored by swept validation F_1. The top five trials by validation F_1 are retrained for 75 epochs at five seeds $(2804, 1234, 42, 7, 13)$, and the winner is chosen by mean threshold-swept validation F_1 across seeds *before any test metric is inspected.* The preregistered tie-breaker, applied when trial means fall within 0.001 validation F_1, prefers lower batch size, then lower learning rate, then lower trial index. The selected base U-Net is trial 44 (`batch_size`=32, learning rate 2.89×10^{-4}, $\lambda_{\mathrm{BCE}} = 0.655$), with mean validation $F_1 = 0.854$ across the five retrained seeds. The full search space is given in Appendix A.

4.5 Threshold selection

A precision–recall curve is swept on the validation set to select the F_1-optimal threshold, which is applied unchanged to the test evaluation; each reported seed uses its own validation-selected threshold, with 0.45 for the canonical seed 2804 (the paper fixes 0.58). Within each run, the saved best epoch is selected by validation Dice at the fixed training threshold, and the saved checkpoint is then threshold-swept for trial ranking and operating-point selection; checkpointing and operating-point selection therefore use related, but different, validation criteria.

4.6 Hough post-processing

A probabilistic Hough transform [17] (`cv2.HoughLinesP`) is applied to a lower-threshold probability canvas to recover faint, linear detections below the segmentation threshold. For each test image the per-patch sigmoid probability maps are max-pooled into a partial canvas covering the sampled test patches rather than the full frame, since the pre-built test set retains a 1:3 patch sample. The canvas is thresholded at $t_{\mathrm{Hough}} = 0.1$, after which `HoughLinesP` is run with a vote threshold of 50, a minimum line length of 100 px, and a maximum gap of 250 px, matching the released ASTA reference implementation of Stoppa et al. [28]. Detected segments are drawn back onto the canvas at 3 px thickness. The evaluation is therefore a patch-canvas approximation to the paper's dense full-image step: all positive patches are retained, so trail-pixel recall remains meaningful, but dropped negatives can underrepresent the background false-positive field.

4.7 Patch CNN classifier for the two-stage variant

The two-stage variant of Section 5.7.1 places a lightweight convolutional neural network (CNN) patch classifier in front of the U-Net. The classifier is intentionally simple: a four-block strided encoder followed by global average pooling and a single fully connected logit head, trained on the sampled training manifest with labels derived from the patch positive-pixel fraction and a `pos_weight` equal to the train-split negative-to-positive ratio. Validation reports both the maximum-F_1 and a high-recall (recall ≥ 0.99) operating point, because a patch rejected by the gate is unrecoverable downstream. The classifier and the U-Net are trained independently and composed only at inference, keeping the gate a single-variable change to the locked pipeline; the classifier never sees test data during U-Net selection.

4.8 Evaluation metrics

Segmentation quality is reported with pixel precision, recall, Dice (equal to F_1 for binary masks), and IoU. Topology is assessed with the centreline Dice (clDice) [26]. Boundary disagreement is isolated with a boundary-tolerant precision/recall/F_1 score at

$k \in \{0, 1, 2, 3\}$ px, where $k = 0$ is the strict metric and $k > 0$ allows a predicted pixel and a ground-truth pixel to match within a small spatial tolerance. The spatial structure of false positives is summarised by their distance to the nearest ground-truth trail pixel, with whole-patch false positives reported separately because they have no defined trail distance. Uncertainty is quantified primarily by a cluster bootstrap over source images, resampling images with replacement and aggregating their patches before recomputing each metric; a patch-level bootstrap is reported only as a secondary diagnostic.

4.9 Determinism and reproducibility

All experiments seed the Python, NumPy, and PyTorch generators (canonical seed 2804), enable cuDNN deterministic mode, fix the cuBLAS workspace, and seed DataLoader workers. Residual GPU non-determinism is absorbed by five-seed retraining of the top trials and seed-averaged reporting.

4.10 Software engineering

The code is an installable Python package with command-line entry points for each pipeline stage. Experiments use YAML configuration plus runtime overrides, so runs are reproducible from configuration and seed.

Correctness is checked by a `pytest` suite covering image and mask pairing, patch geometry, the recovered architecture, loss and segmentation metrics, the shared Hough runner, and training entry points; Ruff handles static checks. Separate CUDA and CPU requirement files are maintained, and a portable CPU Docker image runs the tests by default. NumPy-style docstrings are rendered with Sphinx and served on ReadTheDocs; routines are referenced by file, for example `src/training/train_unet.py`.

Chapter 5

Replication and Error Diagnosis

5.1 Replication of the ASTA detector

The validation-selected winner (trial 44; Section 5.2) is evaluated on the held-out test images at its selected threshold of 0.45. On the sampled test set (1:3 positive:negative) the single-seed model reaches precision 0.852, recall 0.918, and Dice 0.884, with a five-seed mean of precision 0.847 and recall 0.923 (Table 5.1, Figure 5.1). On the parity test set, precision falls to 0.780 at unchanged recall (0.918), as expected once the negative base rate is restored. Recall is therefore reproduced at close to the paper's level, while strict precision sits below the paper's 0.94. The rest of the chapter analyses where that gap comes from, not treating it as an optimisation target.

Table 5.1: Replication comparison against Stoppa et al. (2024). Paper metrics are at threshold 0.58 on a 20,000-patch test set; this work uses the realised 122/24/32 image split and the validation-selected threshold 0.45 (per-seed 0.41–0.50). Cluster-bootstrap intervals are in Section 5.4.

Run	Threshold	Precision	Recall	Dice	IoU
Stoppa et al. (2024)	0.58	0.94	0.94	—	—
This work, sampled (single seed)	0.45	0.852	0.918	0.884	0.791
This work, sampled (5-seed mean)	0.41–0.50	0.847	0.923	0.884	0.792
This work, parity (single seed)	0.45	0.780	0.918	0.843	0.729

5.2 Validation-locked model selection

Trial 44 was selected by mean validation F_1 across the five-seed retraining of the top five trials, before any test metric was inspected (Section 4.4; Figure 5.3). Its validation precision–recall sweep sets the operating threshold of 0.45, carried unchanged into the canonical seed-2804 test evaluation. Across the five reported seeds, the validation-selected

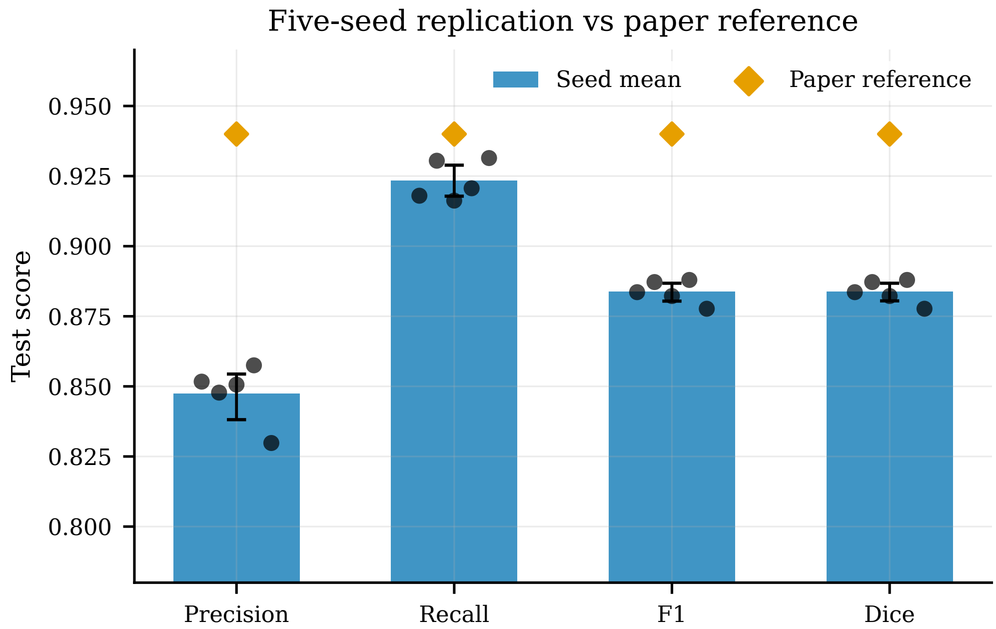

Figure 5.1: Five-seed replication summary against the paper reference, showing precision, recall, F_1, and Dice for the locked validation-selected U-Net configuration.

thresholds range from 0.41 to 0.50, and each seed is evaluated at its own validation-selected threshold. Figure 5.2 shows the validation sweep for the canonical seed; the F_1 plateau is flat between roughly 0.4 and 0.6, so the operating point is not finely tuned.

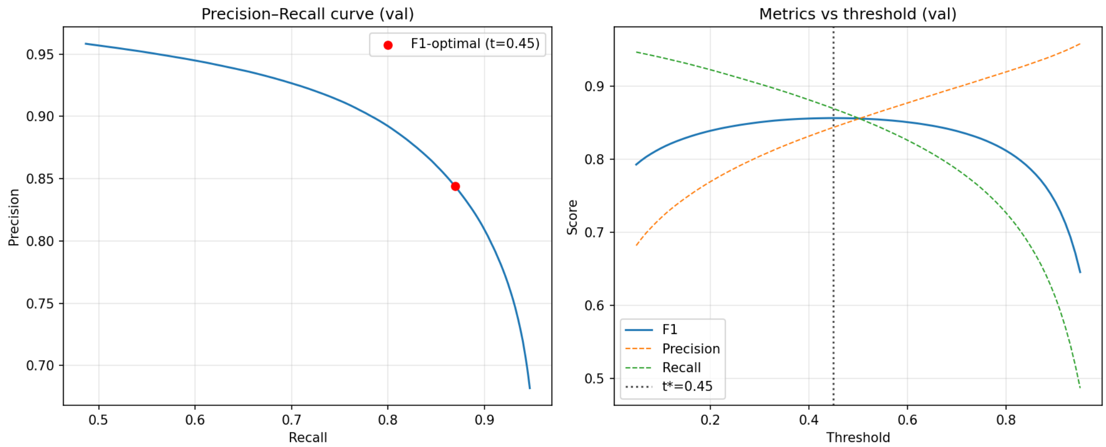

Figure 5.2: Validation sweep for the canonical seed-2804 winner: PR curve with the F_1-optimal point ($t^* = 0.45$, left) and metrics versus threshold (right); the flat F_1 plateau spans the paper's fixed 0.58.

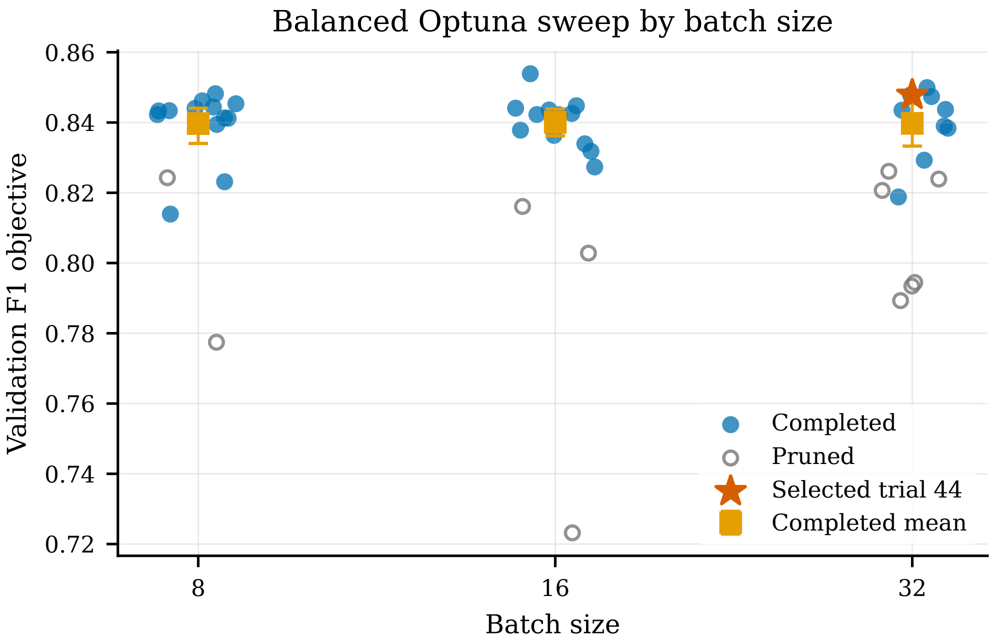

Figure 5.3: Balanced Optuna sweep and validation-only winner selection; pruned trials are hollow markers and the selected trial 44 is starred.

5.3 Hough improves pixel completeness

Probabilistic Hough post-processing of the reconstructed patch canvas raises sampled-support trail-pixel recall from 0.918 to 0.990 for the canonical seed, equivalently reducing the pixel-level false-negative rate from 0.082 to 0.0105 (five-seed mean 0.077 $\rightarrow$ 0.0099). This pixel-completeness rate is the quantity comparable to the paper's reported 6.98 % $\rightarrow$ 3.38 %. The image-level detection false-negative rate is degenerate here: every positive test image is already hit by at least one predicted pixel, so it is 0 before and after Hough and is not comparable to the paper. At patch level, the rate falls only from 0.122 to 0.103, since this stricter diagnostic counts trail-bearing patches with no recovered pixel. Because the canvas is assembled from sampled test patches, all Hough comparisons are patch-canvas approximations to the paper's dense-image setting.

A qualitative post-hoc example (Figure 5.4) shows the mechanism: the U-Net misses a trail stretch over a bright star, and Hough recovers ground-truth pixels that appear only after the geometric stage. The example is a mechanism illustration, not a benchmarked bridge-rate estimate.

5.4 Uncertainty quantification

Cluster-bootstrap 95 % intervals over the 32 test images (1,000 resamples, single-seed winner) are precision 0.852 [0.806, 0.891], recall 0.918 [0.888, 0.946], Dice 0.884 [0.857, 0.904],

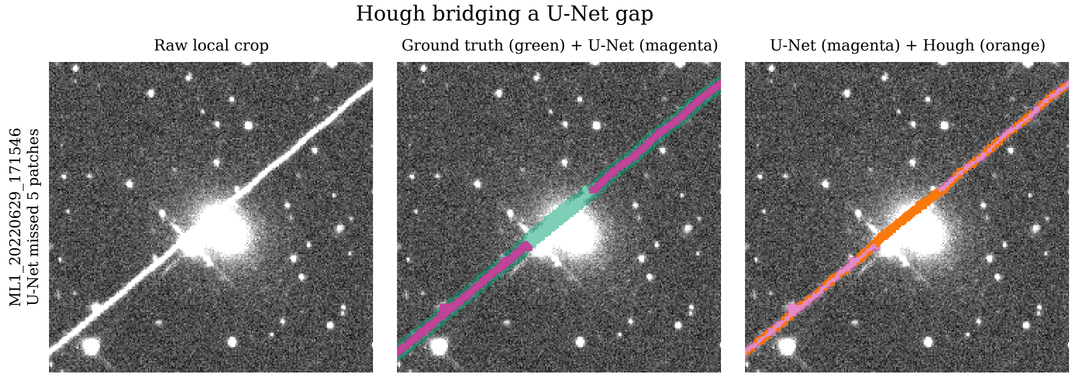

Figure 5.4: Hough bridging a U-Net gap. Panels: raw image; ground truth (green) with U-Net prediction (magenta); U-Net (magenta) with Hough bridge (orange). Ground-truth pixels over the bright star are recovered only by Hough.

and IoU 0.791 [0.749, 0.825]. These image-level intervals are several times wider than the patch-level bootstrap (Dice [0.876, 0.892]), which is reported only as a secondary diagnostic; the 32 source images are the independent units. Across the five trial-44 seeds, the sampled-test means and standard deviations are precision 0.847 ± 0.009, recall 0.923 ± 0.006, and Dice 0.884 ± 0.004, so seed variation is smaller than the image-cluster uncertainty but still visible.

5.5 Qualitative prediction checks

A representative selection of locked-winner test predictions is shown in Figure 5.5; false-positive galleries for the winning U-Net and attention variant on the same eight patches are in Appendix B. They include diffraction-spike examples: low-Dice false positives on stellar structure and high-Dice cases where bright stars are correctly ignored.

As a final qualitative check, the locked model was applied cold to nine predeclared DECam frames with RECA-measured streaks (Figure 5.6). With no masks, no metric is reported; on author review, eight of nine visible streaks are fully traced, while the very faint full-frame SORCE streak is traced along only about a tenth of its length and counted as a miss. This sits at the low-contrast end predicted by Section 5.6.4, illustrating out-of-domain behaviour rather than a cross-dataset benchmark.

5.6 The strict precision gap is boundary-scale

The diagnostics below give a consistent pattern: most false-positive pixels lie close to labelled trails, consistent with thin or imperfect mask boundaries, while a smaller whole-patch component remains genuine background confusion. The interpretation stays provisional: no independent re-annotation set is available.

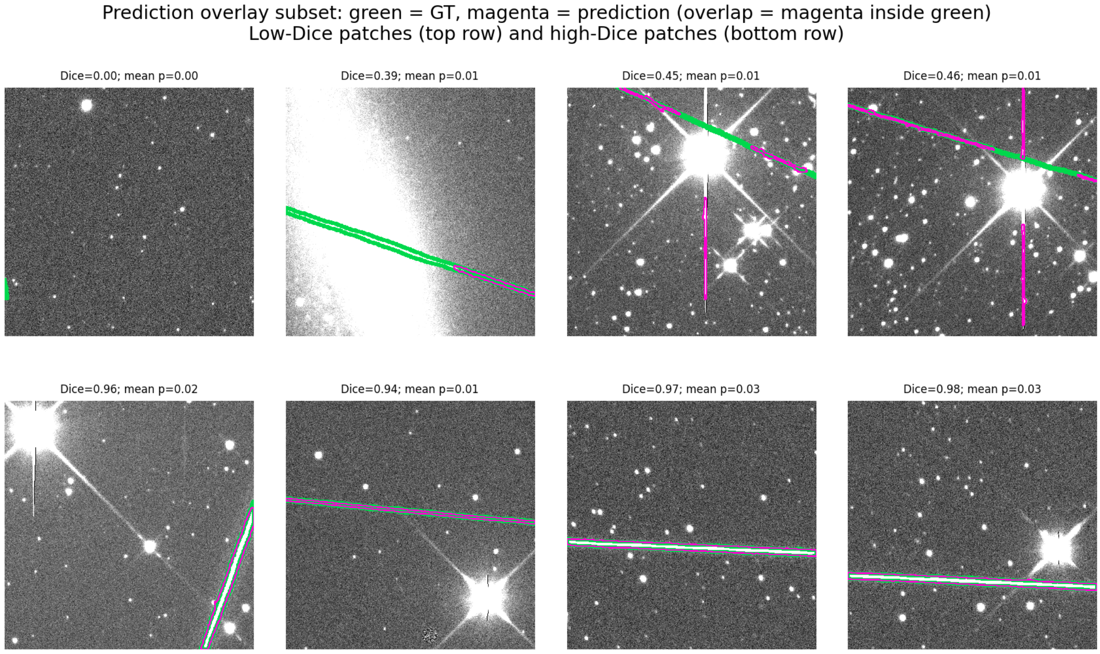

Figure 5.5: Representative test-patch predictions for the locked winner: image, ground-truth mask (green), and predicted mask (magenta), with overlap shown as magenta inside the green band.

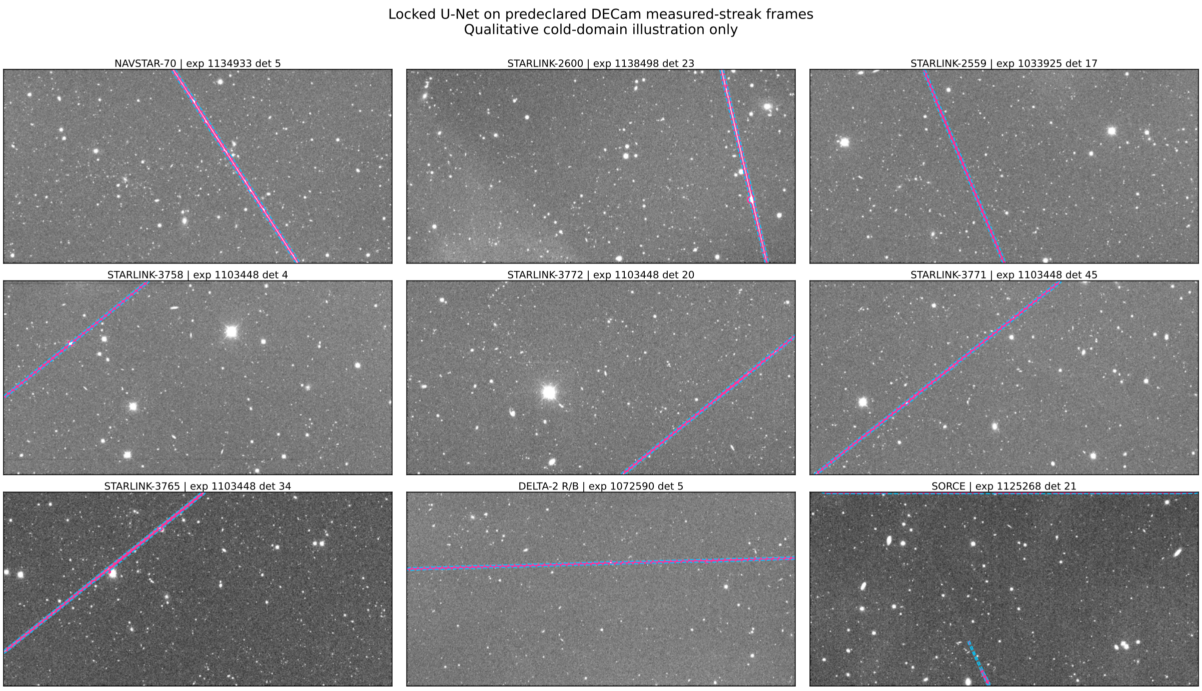

Figure 5.6: Locked MeerLICHT-trained U-Net applied cold to predeclared DECam measured-streak frames (panels rotated for layout; magenta: U-Net contour at $p \geq 0.45$; cyan dashed: Hough). Qualitative illustration only: no ground-truth masks or benchmark metrics exist.

5.6.1 False-positive decomposition

Decomposing the false-positive pixels of the winning model shows that only 17.5 % are inter-patch (the network lighting up in trail-free patches), while 82.5 % are intra-patch (over-painting inside trail-bearing patches) (Table 5.2). A patch-level gate addresses only the inter-patch share, so this 17.5 % is a hard upper bound for such a gate; hard-negative mining retrains and is only loosely bounded by it.

Table 5.2: Decomposition of false-positive pixels. Only the inter-patch component is directly removable by a patch-level gate (a looser reference for hard-negative mining).

Component	Pixels	Share
Intra-patch FP	300,745	82.5%
Inter-patch FP	63,662	17.5%

5.6.2 Boundary-tolerant evaluation

Re-scoring under a symmetric boundary tolerance of $k \in \{0, 1, 2, 3\}$ px (RECT structuring element; a predicted pixel within k px of ground truth counts towards precision, and a ground-truth pixel within k px of a prediction counts towards recall) isolates boundary disagreement from genuine misses (Figure 5.7). Allowing a single pixel of tolerance, the five-seed mean precision rises from 0.847 (strict) to 0.946, recall from 0.923 to 0.980, and the ± 1 px F_1 reaches 0.962: the precision gap to the paper largely closes under a one-pixel tolerance. A one-sided check separates this from a recall-for-precision trade: re-scoring the same predictions against ground truth dilated by one pixel raises precision from 0.852 to 0.937 but drops recall to 0.793, whereas the symmetric tolerance raises both. This is consistent with residual error dominated by boundary placement rather than spurious detection. The claim remains provisional without a gold-standard re-annotation set, and the tolerant score is not treated as equivalent to the paper's strict 0.94/0.94.

5.6.3 False-positive distance to mask

The distance from each false-positive pixel to the nearest ground-truth trail pixel concentrates at small values (Figure 5.8). Of all false positives, 57 % lie within 1 px; of the distance-defined false positives (those in trail-bearing patches, excluding the $\sim$17 % whole-patch false positives that have no defined distance), 69 % lie within 1 px, with a median of 1.6 px. Both denominators are reported because the 69 % figure alone overstates the global picture. The distribution is consistent with a boundary-placement-dominated residual error plus a bounded genuine-background minority.

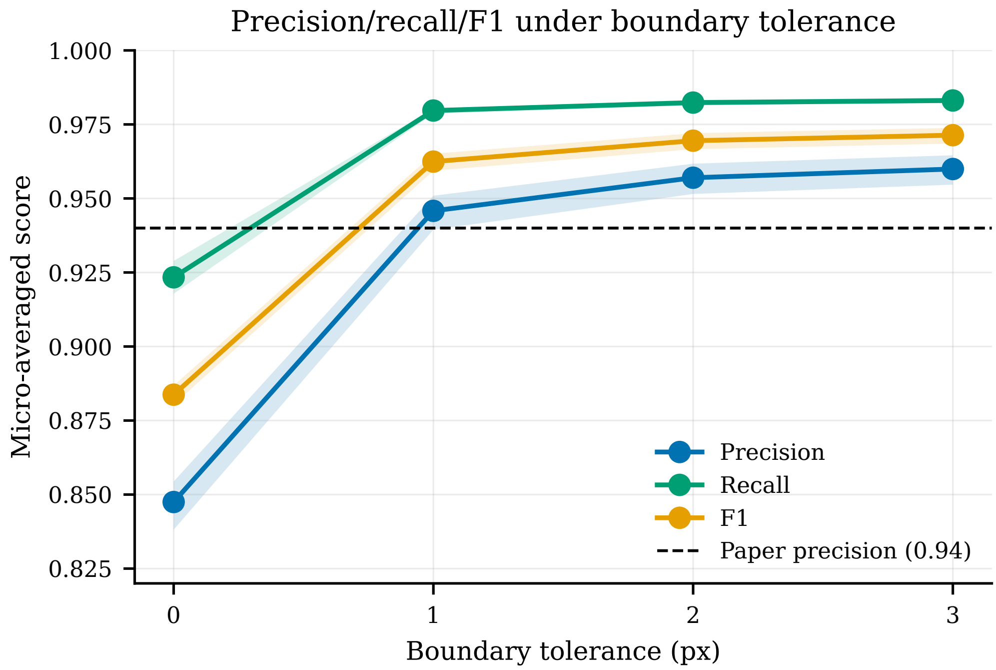

Figure 5.7: Boundary-tolerant metrics versus tolerance k (micro-averaged); the horizontal line marks the paper's strict precision of 0.94. One pixel of tolerance largely closes the strict gap, but the relaxed score is not directly comparable to the paper's strict metric.

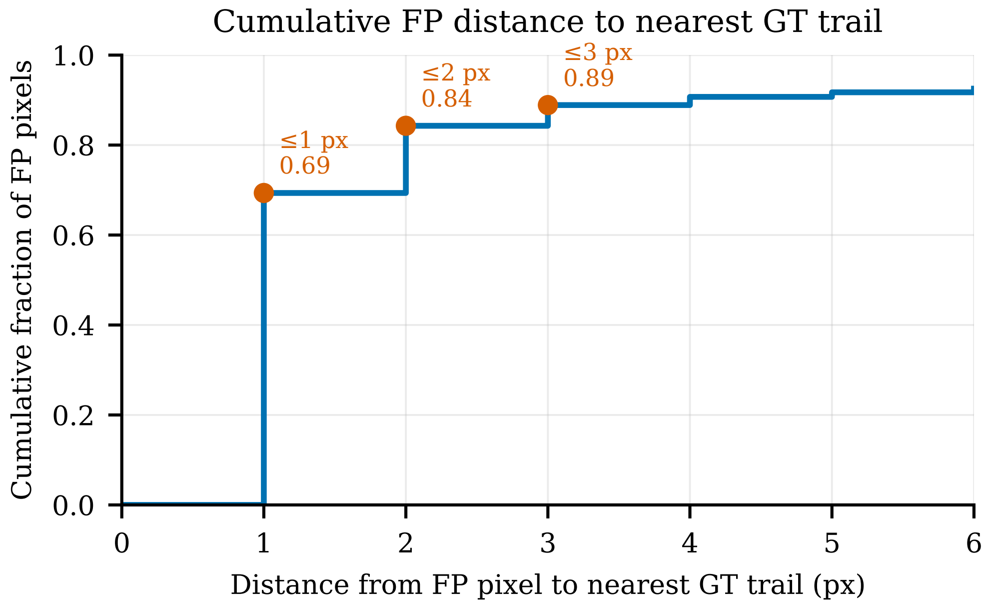

Figure 5.8: Cumulative fraction of false-positive pixels within distance d of the nearest ground-truth trail. The distance-defined curve excludes 63,662 whole-patch FP pixels; 69.3 % of distance-defined and 57.2 % of all FP pixels lie within 1 px.

5.6.4 Relative display-contrast diagnostics

The locked winner was rescored post hoc on the 31 positive test images using raw 8-bit PNG intensities as a *relative display-contrast proxy*, not calibrated flux or signal-to-noise (Appendix Figures B.1–B.3); median cross-sections through labelled trails make the proxy concrete (Figure B.2), and fitting calibrated flux in an aperture would need counts the 8-bit renders do not carry (Section 2.1). Across 55 cleaned trail components, the low-contrast tertile has the lowest pre-Hough recall (0.853) and the largest Hough-added ground-truth fraction (0.092), with post-Hough recall at least 0.945 in every tier, so Hough acts exactly where the network evidence is weakest. Near-trail false positives sit at intermediate display excess between background and labelled-trail pixels (Figure B.3): consistent with boundary under-capture near labels and trail-like background further away, and descriptive rather than evidence of label error.

5.6.5 Topology-aware comparison (clDice)

The centreline Dice (clDice) tracks pixel Dice and preserves the architecture ordering: 0.847 for the base model versus 0.838 for the attention variant (-0.009), against pixel Dice 0.884 and 0.879 (-0.005). Topology therefore agrees with the null attention result and shows no hidden connectivity advantage.

5.7 Model variants are bounded by the error decomposition

The model variants test the error diagnosis. If the precision gap were dominated by empty-patch false positives, a patch classifier or hard-negative training should give a material precision gain; if by missing faint structure, attention gates might improve overlap or topology. The observed results are narrower: the gate shifts precision in the direction predicted by the 17.5 % inter-patch bound, and attention does not improve the base architecture.

5.7.1 Classifier-gated detector

The classifier-gated detector (gate architecture and training in Section 4.7) is reported descriptively as an operating-characteristic surface over the classifier and U-Net thresholds (Figure 5.9), not as a single selected replacement for the locked U-Net. The gate suppresses only empty-patch false positives, so its gain is bounded by the inter-patch share. At the illustrative high-recall operating point used for Figure 5.9 ($t_{\text{clf}} = 0.18$, $t_{\text{U-Net}} = 0.45$), the gate routes 2,625 patches to the U-Net (75.3 % of the sampled test set), improving precision from 0.852 to 0.864 while reducing recall from 0.918 to 0.914 and giving Dice 0.888. End-to-end recall counts classifier-rejected positive patches as false negatives, exposing the classifier false-negative rate (0.096 at this operating point) as a hard recall ceiling. The gate gives a bounded precision gain and a real compute saving at this operating point, but not an order-of-magnitude saving: it skips 24.7 % of sampled-test U-Net forwards. Larger savings would require a more aggressive gate on a denser, naturally empty-dominated patch stream.

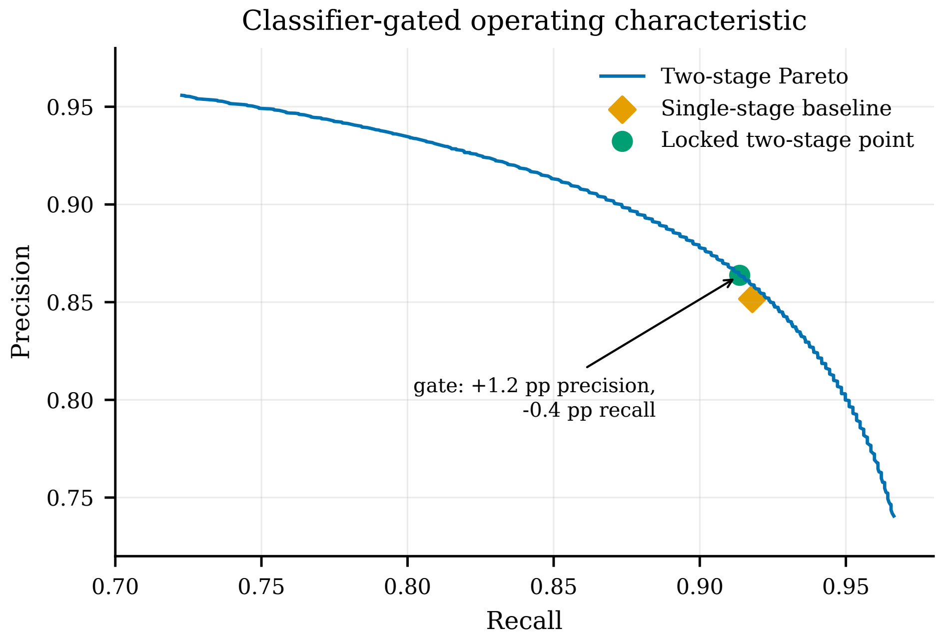

Figure 5.9: Two-stage operating-characteristic (Pareto) surface over the classifier and U-Net thresholds; test-side and descriptive, not a validated improvement. The annotated point shifts precision +1.2 pp and recall −0.4 pp versus the locked baseline.

5.7.2 Attention-gated U-Net

Under a matched protocol (the balanced sweep, top-five five-seed retraining, and validation-only selection of Section 4.4), the attention-gated U-Net [22] does not improve on the base network despite carrying roughly 1 % more parameters. Its winner (trial 7) reaches test precision 0.838, recall 0.924, and Dice 0.879 at threshold 0.48, against the base Dice of 0.884 (Table 5.3). The five-seed validation-restudy means are 0.854 for the base model (90 % interval [0.851, 0.856]) and 0.851 for the attention model ([0.850, 0.852]), so the comparison is best read as matched-protocol null evidence rather than as a decisive architectural ranking. Together with the clDice result (Section 5.6.5), the attention variant reveals no hidden connectivity or faint-trail advantage on this dataset.

Table 5.3: Five-seed test means for the base and attention-gated U-Nets.

Metric	Base U-Net mean	Attention U-Net mean	Delta
Precision	0.847	0.848	+0.001
Recall	0.923	0.919	-0.005
F_1	0.884	0.882	-0.002
Dice	0.884	0.882	-0.002

5.8 Training and inference sensitivity

5.8.1 Data efficiency

Retraining the winning configuration on nested 30/50/70/100 % subsets of the training images (validation and test held fixed) shows validation Dice flat to 50 %, rising at 70 %, then near-constant (Figure 5.10). Because the hyperparameters are transferred from the 100 %-tuned winner rather than re-tuned per fraction, the curve is a lower bound at each data size; it is a single-seed, indicative result, plotted without a confidence band. The fixed threshold-0.5 validation Dice values are 0.842, 0.842, 0.854, and 0.856 for 30, 50, 70, and 100 % of the training images, respectively, so the main gain occurs before the final fraction. On this single-seed curve the plateau suggests that the 178-image subset does not obviously bind this architecture, which is consistent with the replication reading of Section 5.1.

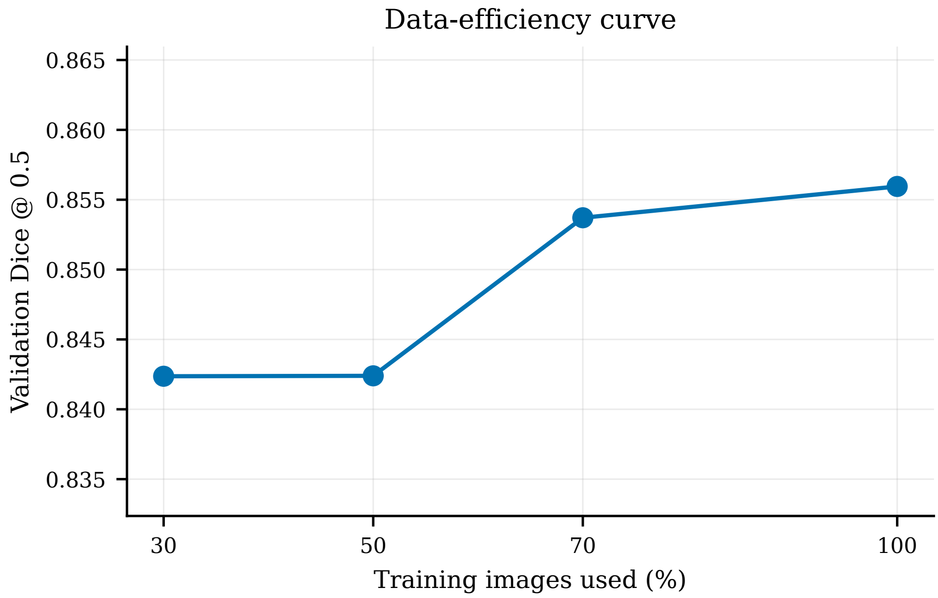

Figure 5.10: Validation Dice versus training-image fraction (single seed 2804; fixed-threshold history metric at 0.5, not threshold-swept). Hyperparameters are transferred from the 100 % winner rather than re-tuned per fraction.

5.8.2 Hard-negative mining

Appending 150 mined hard-negative (false-positive empty) patches to the training set yields a small precision gain in this single-seed confirmation: +0.007 precision ($0.852 \rightarrow 0.859$) at essentially unchanged recall (-0.001), for +0.003 Dice. The gain is bounded by, and consistent with, the 17.5 % inter-patch false-positive share.

5.8.3 Soft-label (dilated-target) pilot

A single-seed pilot replacing the hard targets with 1 px dilated soft-edged targets (training only; validation and test masks unchanged) improves no overlap, precision or topology metric (Table 5.4). The soft model predicts wider trails, with its validation-optimal threshold rising from 0.45 to 0.62, giving recall +0.008 but precision -0.022, exact Dice -0.009, clDice -0.013, and a tied ± 1 px F_1: the boundary band is absorbed into wider predictions rather than sharpening localisation. The pilot is reported as a null result, consistent with the boundary-placement interpretation.

Table 5.4: Soft-label (dilated-target) pilot versus the hard-label winner; a single-seed protocol variant with no metric gain. Both columns are canonical seed-2804 values, so the hard-label ± 1 px F_1 here is the seed-2804 figure (0.963); the five-seed mean is 0.962 (Section 5.6.2).

Metric	Hard-label winner	Dilated-soft pilot	Delta
Exact Dice	0.884	0.875	-0.009
± 1 px F_1	0.963	0.962	-0.001
clDice	0.847	0.835	-0.013

5.8.4 Ensembling with test-time augmentation

A five-checkpoint ensemble with test-time augmentation reaches Dice 0.8885 (precision 0.852, recall 0.929 at threshold 0.44), against the single-model 0.8836. The +0.005 Dice improvement is comparable to the five-seed dispersion of the single model, so it is not treated as a material gain.

5.8.5 Probability-stratified Hough

Replacing the binary Hough input with a probability-stratified canvas is a clean null: pixel recall changes by +0.00003, the patch-level false-negative rate is identical, and Hough false-positive pixels rise from 2,725,702 to 2,785,274. The locked binary Hough already recovers essentially all recoverable pixels, so stratification only adds false positives.

Chapter 6

Discussion

6.1 Deviations from the target paper

Faithful replication is a matter of degree, so Table 6.1 makes the remaining departures from Stoppa et al. [28] explicit. They are forced choices where the paper omits training hyperparameters; deliberate methodology choices such as image-level stratified splits, a validation-swept threshold, and sampled-plus-parity testing; and implementation conventions such as the PyTorch port and logits-with-external sigmoid output. Alignment only in spirit, as with the squared-denominator Dice and stride-528 tiling, is not described as paper-faithful.

Three deviations change the headline comparison: the swept threshold and different test population confound precision, leaving recall as the closest like-for-like quantity (Section 5.1); the patch-canvas Hough makes completeness figures approximations to the paper's dense-image setting (Section 5.3). The remaining departures were not individually isolated: architecture tests lock the framework port and output convention, image-level splitting removes patch-level leakage, and calibrated noise was validation-screened before adoption.

6.2 Interpreting the headline comparison

Section 6.1 leaves recall as the like-for-like quantity, and even that spans different populations: five-seed recall 0.923 ± 0.006 against the paper's 0.94 is measured on 3,488 sampled patches (12,800 at parity) from 32 images, against the paper's 20,000 of unreported class balance. For precision the prevalence confound is measured, not assumed: the fall from 0.852 to 0.780 at identical recall and a fixed threshold is the direct size of the base-rate effect, while the annotation confound is localised at about one pixel by Section 5.6. Evaluating at the paper's fixed 0.58 rather than the swept 0.45 raises precision by only 0.025 on both test sets, so the threshold is a minor contributor. A single headline precision figure

Table 6.1: Documented deviations from Stoppa et al. (2024).

Aspect	This work	Target paper
Framework	PyTorch	Keras / TensorFlow
Architecture source	Reconstructed from the released Keras model configuration and reimplemented in PyTorch	Keras model used by the paper
Output convention	Logits; sigmoid applied in loss/evaluation	Keras model includes a final sigmoid
Dice formulation	Squared-denominator Dice, following the released ASTA code	Standard Dice in the paper text
Patch tiling	528×528 at stride 528 (exact 20×20 grid)	528×528 input; tiling not specified
Threshold	Validation-set PR sweep (F_1-optimal)	Fixed at 0.58
Splits	Image-level, stratified by trail-pixel quartile; realised 122/24/32	Not specified at image level
Test sampling	Sampled test at 1:3 positive:negative; separate parity test retains all negatives	20,000-patch test set
Hough input	Patch-reconstructed canvas; parity accounting includes all test-image false positives	Dense full-image prediction
Hyperparameters	Validation-only Optuna search (LR, BCE/Dice weighting, batch-size regime)	Not fully published
Noise augmentation	Calibrated train-only signal-dependent noise	Not specified

therefore mixes detector behaviour with population and labelling differences it cannot separate.

The Hough comparison is favourable for completeness but not for strict precision. The pixel-level false-negative rate falls from 0.082 to 0.0105, an approximately 87 % reduction against the paper's 51 % ($0.0698 \rightarrow 0.0338$): comparable before the geometric stage, stronger after. The same run raises strict false-positive area, cutting strict pixel precision from 0.780 to 0.410, partly because the 3 px Hough lines extend beyond the labelled trail extent; with the canvas reconstructed from patches rather than dense full-frame inference (Section 4.6), this is comparable completeness, not superiority. The same discipline applies to the tolerant scores: the ± 1 px precision 0.946 and recall 0.980 locate the residual at the boundary scale, but a relaxed metric is never set against the paper's strict 0.94/0.94.

6.3 Limitations

The findings above are bounded by seven limitations.

- **Mask undercapture.** The annotations occasionally undercover faint endpoints and low-contrast stretches (Section 2.5). The effect is asymmetric: a correct detection of an unlabelled trail pixel scores as a false positive, and a model trained on such labels also learns to suppress faint extensions. Label-referenced precision is therefore a lower bound where the labels undercover. The re-annotation audit of Section 7.2 is the direct mitigation.

- **Dataset size.** The subset holds 178 images, of which 32 form the test set. Image-level bootstrap intervals are accordingly several times wider than patch-level ones (Section 5.4), the image-level detection rate saturates, and the protocol-sensitivity results (data efficiency, hard negatives, soft labels) are single-seed pilots.

- **Sampled evaluation distribution.** The sampled test set keeps a 1:3 positive-to-negative patch prior, which inflates precision relative to the natural, roughly 93 %-empty population; the parity evaluation restores the negatives but is still scored at the sampled-validation threshold. The sampled-to-parity precision drop (0.852 to 0.780) bounds the size of this bias; recall is unaffected because all positive patches are retained.

- **Patch-canvas Hough.** Hough runs on canvases reconstructed from patch predictions, not on native dense full-frame inference. The parity evaluation restores all test patches for false-positive accounting, but the post-Hough precision remains sensitive to line rasterisation thickness and patch-canvas reconstruction. The pixel-completeness comparison remains meaningful because every positive patch is retained.

- **Single telescope.** All training and quantitative evaluation use MeerLICHT data; transfer to another instrument's pixel scale, point-spread function, or background is untested, and the DECam application (Figure 5.6) is qualitative only.

- **PNG-only inputs.** The 8-bit renders cap the dynamic range and carry no WCS, exposure, or photometric headers. Trails near the display floor may be undetectable in principle, and the contrast statements of Section 5.6.4 are display-space proxies. The direction of any bias is unclear, because the labels were drawn on the same renders.

- **Single-band imagery.** Each frame provides one display channel, so no claim extends to multi-band trail morphology [37].

6.4 Scientific implications

The error characterisation carries three implications beyond this replication, and a practical reading for deployment.

First, strict precision measured against incomplete labels is a lower bound, and the Hough stage makes this concrete. The Hough step is designed to extrapolate linear extent that the network under-detects; where the annotation also undercovers that extent, the recovered pixels score as false positives. The pipeline is penalised by the metric for exactly the recovery it was built to perform, so post-Hough strict precision understates the true masking quality. Reported precision for this family of detectors should be read jointly with a statement of label quality, and comparisons across papers with different annotation conventions conflate detector quality with labelling convention.

Second, pixel-level false-negative rates have an instance-level blind spot. The semantic formulation labels every trail pixel identically, so two near-parallel trails merged into a single detection lose nothing under any pixel metric reported here. Freshly launched constellation "trains" produce precisely this geometry, and the failure would matter wherever downstream consumers need trail counts or per-satellite association rather than masks.

Third, the boundary-placement finding suggests a practical reporting standard. The dominant residual sits within about one pixel of the labelled boundary, and no tested capacity, protocol, or inference change moves it materially. This is consistent with the strict score being saturated by annotation granularity on thin structures — although, without a re-annotation audit, label thickness and a systematic one-pixel over-paint cannot be distinguished. If that reading holds, annotation protocols should fix and record a mask-width convention, and evaluations of thin-artefact detectors should report a $\pm 1\,\mathrm{px}$ boundary-tolerant operating point alongside the strict score, so that boundary convention and detection failure are separated rather than summed.

In practice, the decomposition reads as deployment guidance. Pipelines that dilate detected masks before stacking would be less sensitive to a residual confined within about one pixel of the boundary, so the strict-precision gap may matter less for masking, though this was not tested. The classifier gate is best treated as a compute feature: it skips 24.7 % of U-Net forwards on the sampled test set, and would skip more at the natural ∼93 %-empty prevalence (not evaluated); its precision headroom is capped by the 17.5 % inter-patch share and its false-negative rate sets a hard recall ceiling. Thresholds likewise do not transfer across class prevalence: the sampled-to-parity fall occurs at a fixed threshold, so a deployed detector should re-select its operating point on data at the deployment prevalence.

Chapter 7

Conclusions and Future Work

7.1 Conclusions

This dissertation replicated the U-Net plus probabilistic Hough satellite-trail detector of Stoppa et al. [28] on a 178-image MeerLICHT subset, with the architecture recovered from the released model and the final model and threshold locked on validation data alone. The replication reproduces the paper's recall closely: five-seed sampled-test recall is 0.923 ± 0.006 (single seed 0.918) against the reported 0.94, with Dice 0.884 ± 0.004. The Hough stage reproduces the paper's completeness behaviour, raising trail-pixel recall from 0.918 to 0.990 on the patch canvas, a pixel-level false-negative rate of $0.082 \rightarrow 0.0105$ against the paper's $6.98\,\% \rightarrow 3.38\,\%$. Strict pixel precision sits below the published level, at 0.847 ± 0.009 on the sampled test set and 0.780 (single seed) once the natural negative prevalence is restored at unchanged recall. That fall identifies strict precision as prevalence-dependent and leaves recall as the like-for-like comparison across test populations.

The follow-up analyses localise the precision gap rather than merely reporting it: 82.5 % of false-positive pixels lie inside trail-bearing patches, a one-pixel boundary tolerance raises five-seed precision from 0.847 to 0.946 (F_1 0.962), and 57 % of all false positives lie within one pixel of a labelled trail. The variants behave as that decomposition predicts: the classifier gate shifts precision +1.2 pp, bounded by the 17.5 % inter-patch share; the attention-gated U-Net is a null at matched capacity, with clDice confirming no hidden topology advantage; validation Dice plateaus beyond roughly 70 % of the training images; hard-negative mining adds +0.007 precision; the soft-label pilot improves no overlap, precision or topology metric; and ensembling with test-time augmentation and probability-stratified Hough sit within seed noise. The residual error is bounded by the error decomposition: none of the tested architecture, training-protocol, or inference changes moves it materially, and these bounded and null outcomes are what make the boundary-placement diagnosis credible.

The project also demonstrates a methodology. Faithful replication required recovering the architecture from released weights, preregistering selection rules, and separating sampled from parity evaluation; each step surfaced a discrepancy a looser reproduction would have absorbed into an unexplained shortfall. Treating the remaining gap as an object of analysis then converted a headline deficit into a measured, bounded mechanism consistent with one-pixel boundary disagreement. Conducted at this level of fidelity, replication is not a re-implementation exercise but an instrument for locating where a published result's accuracy actually resides: in the detector, in the labels, or in the metric.

7.2 Future work

Several directions follow from the findings. The most decisive is a *gold-standard re-annotation audit* of a sample of test trails: it would convert the suggestive boundary-placement reading into a measured label-noise floor and establish how much of the strict precision gap is attributable to mask granularity rather than detector error.

A second priority is *cross-dataset generalisation*: the locked model was applied qualitatively to DECam frames (Figure 5.6) for illustration only, and a labelled cross-instrument evaluation, with a classical baseline in the spirit of Serrano Mendoza et al. [25], would test whether the detector transfers. Attention [22] and richer architectures could then be re-examined on a larger multi-telescope corpus, where the matched-capacity null may not persist.

Finally, the pixel-level metric is blind to instance-level multiplicity: freshly launched constellation "trains" of near-parallel streaks would be a useful *synthetic robustness probe*, alongside a Hough length-gate (`minLineLength`) study for short linear artefacts and a resolution-robustness check across instrument pixel scales. Operationally, real-time integration in the reduction pipeline, scoring each frame as it arrives from the telescope and surfacing flagged trails through a simple review interface, and the sky-position and time-trend analyses that calibrated FITS with WCS headers would unlock, would extend the work towards a deployable tool.

Two further directions complete the picture. Evaluation should move beyond pixel metrics to an *instance-level* detection score (object-level precision, recall, and F_1 over matched trail instances), which would expose the merged near-parallel-trail failure mode that pixel-level rates cannot register and would make the constellation-train probe quantitative. Trail *removal* is the natural downstream step: a classical route would combine the detected mask with the fitted Hough geometry to interpolate across the trail from clean flanking pixels, evaluated by synthetic injection into verified-clean patches. On 8-bit renders this is cosmetic reconstruction rather than photometric recovery; production pipelines mask and reject, and a real source under the trail remains the key failure mode.

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Appendix A

Hyperparameter Search Space

Table A.1 records the Optuna search space; all other training choices are fixed by the methodology in Chapter 4.

Table A.1: Optuna hyperparameter search space.

Parameter	Type	Range / values	Notes
`learning_rate`	log-uniform	$[10^{-4}, 10^{-3}]$	Common range for all batch sizes
`bce_weight`	uniform	[0.2, 0.8]	$\lambda_{\text{Dice}} = 1 - \lambda_{\text{BCE}}$
`batch_size`	fixed allocation	$\{8, 16, 32\}$	45 trials total; 15 per batch size

Appendix B

Supplementary Results

B.1 Post-hoc Trail and Error Diagnostics

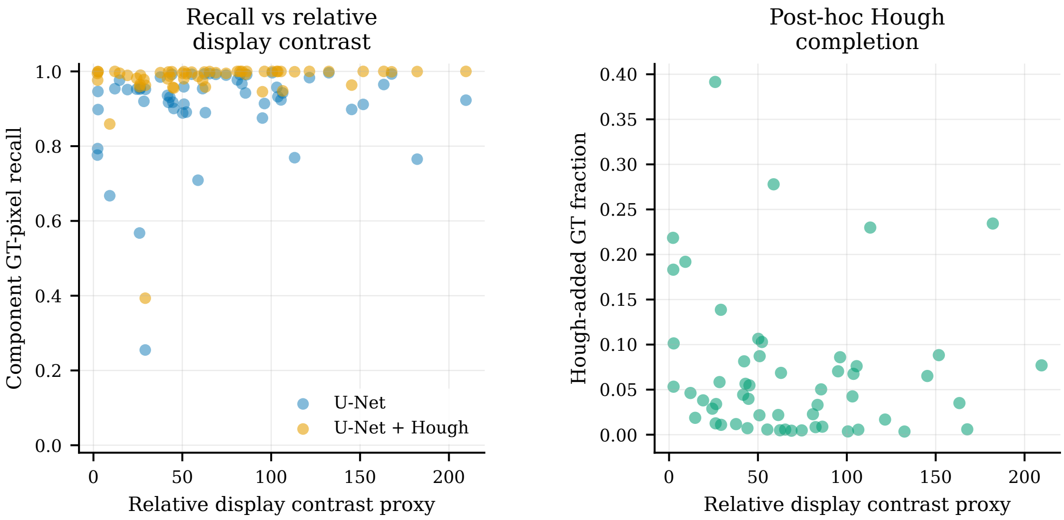

Figure B.1: Relative display contrast versus U-Net recall and Hough completion for the locked winner; the contrast proxy is descriptive, not calibrated flux or SNR.

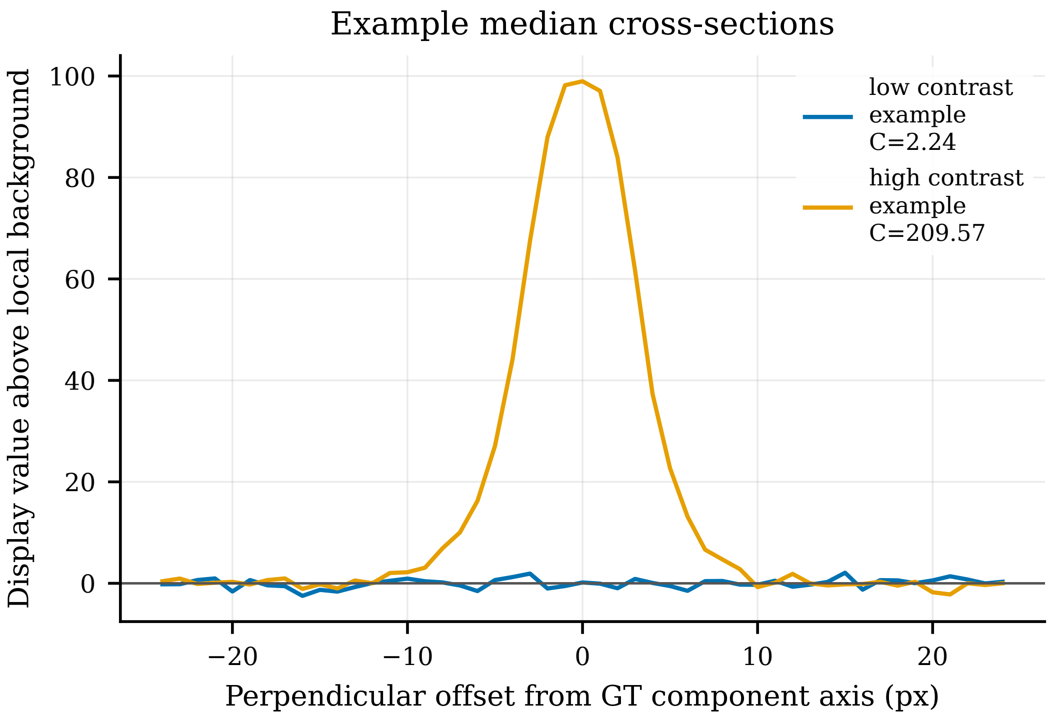

Figure B.2: Median cross-sections through labelled trails in display-space PNG values; descriptive examples of the relative-contrast proxy.

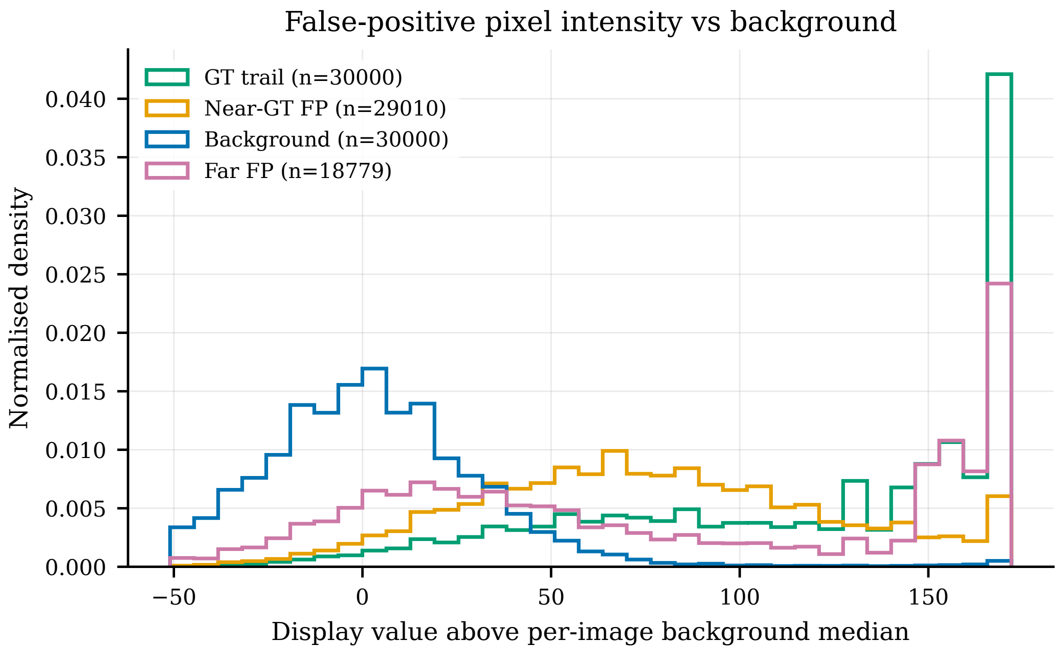

Figure B.3: False-positive pixel display intensity versus labelled trail and background pixels. The KS distances are consistent with boundary under-capture and background confusion, but remain descriptive.

B.2 Prediction Galleries

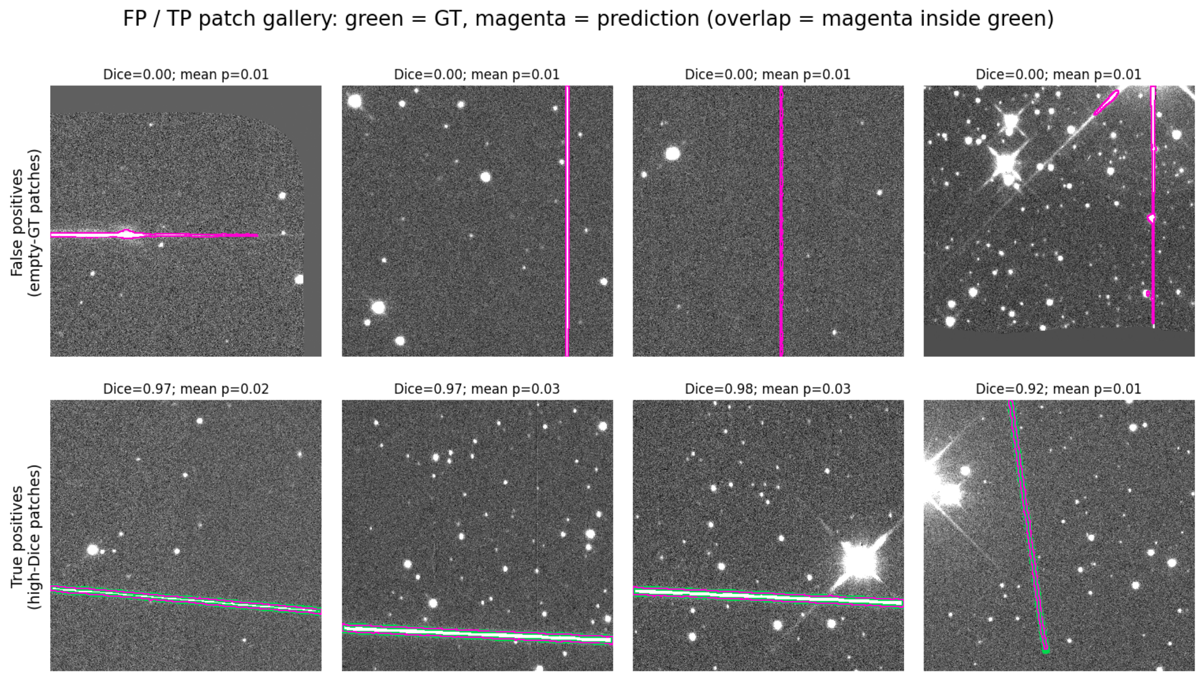

Figure B.4: False-positive and true-positive gallery for the locked winner, including bright-star diffraction-spike examples alongside high-Dice cases where bright stars are correctly ignored.

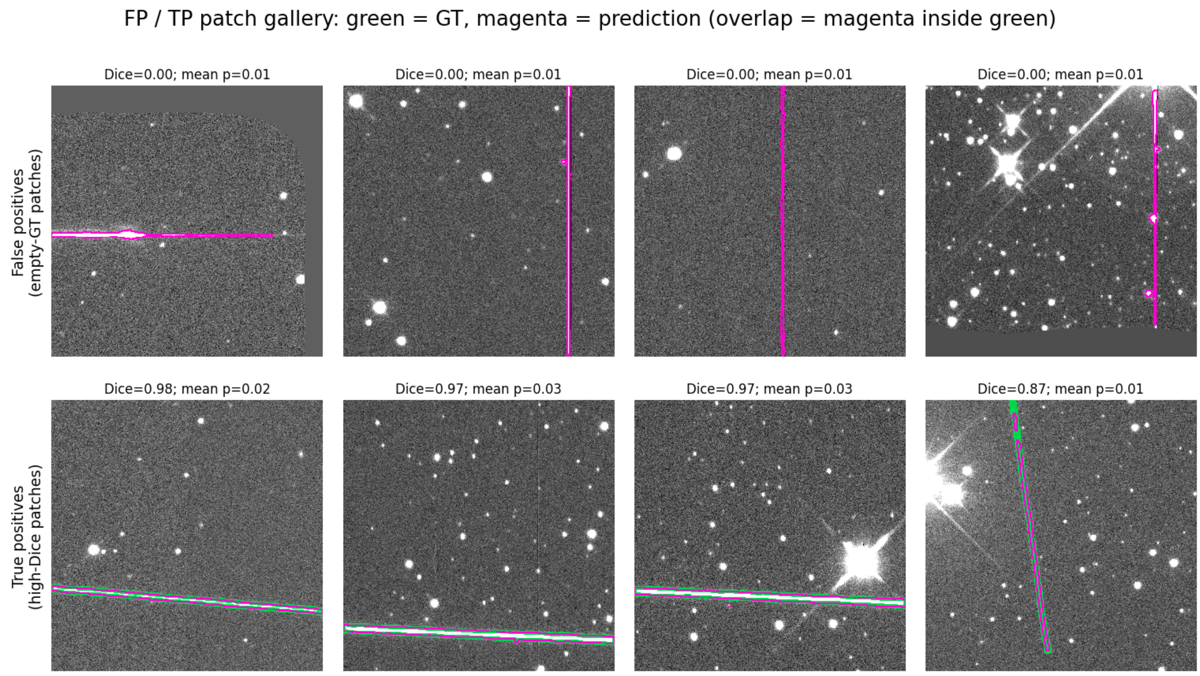

Figure B.5: The same eight patches for the attention-gated U-Net, in the order of Figure B.4, each with that model's own segmentation and Dice score.

Appendix C

Use of Auto-generation Tools

Codex (ChatGPT) and Claude Code were used to support this work, primarily in the following areas: advice and usage support for libraries such as PyTorch and PIL; boilerplate code for plotting, testing, and scripting; function docstrings; and review and audit of the project structure for completeness, consistency, accuracy, and software-engineering best practice. All committed code and all prose were reviewed and manually approved; all reported experimental metrics were checked against the repository artefacts, and external contextual statistics derive from the cited sources.